

Mobile Mapping, Artificial Intelligence and Digital Data Optimize Road Infrastructure Management

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In the realm of geospatial data collection, mobile mapping has emerged as a transformative force, offering significant advantages over traditional surveying methods for a range of applications, including road condition evaluation and construction. As shown in the accompanying figure, Trimble offers a comprehensive AI-driven road condition evaluation and construction workflow, providing the following benefits to municipalities and transportation departments:

Enhanced Efficiency.

Automated data processing reduces the time from data collection to decision-making, enabling quicker responses to maintenance needs.

Improved Safety. Mobile mapping at traffic speeds minimizes the exposure of survey crews to hazardous conditions.

Cost Savings. Automation and efficient data handling reduce labor costs and resource allocation.

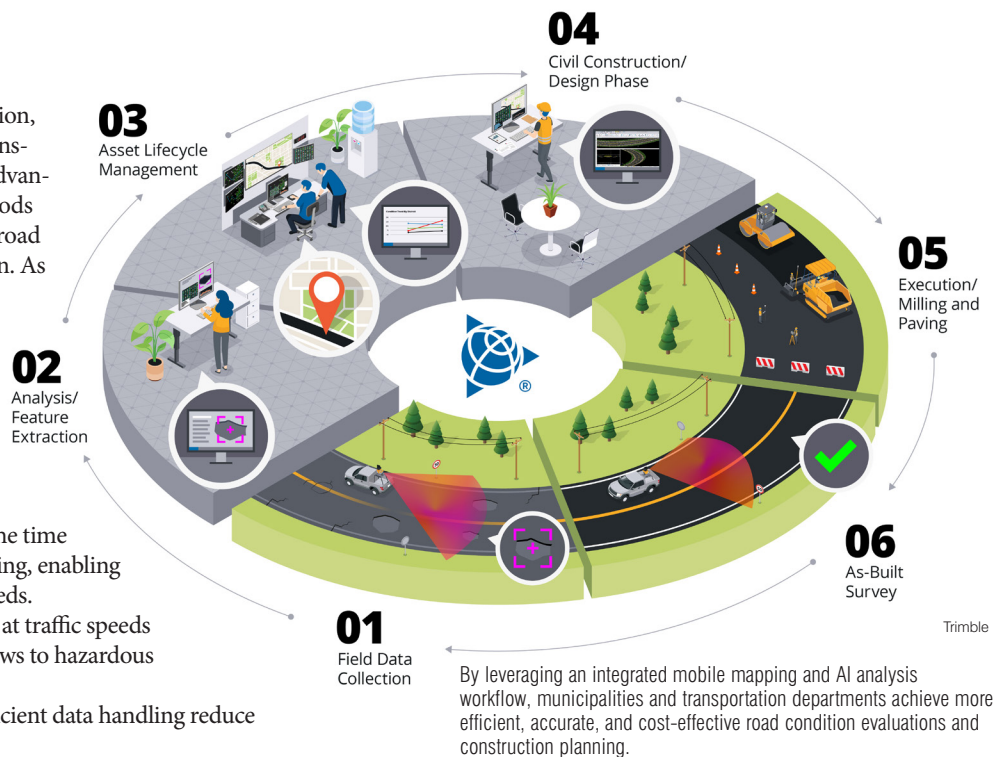
Data Accuracy. High-resolution data ensures precise assessments, supporting better infrastructure planning and maintenance.

Seamless Integration. Compatibility with geographic information system (GIS) platforms such as Esri's ArcGIS allows for easy incorporation into existing workflows.

Workflow Phases

As cities continue to grow around the world, road construction, maintenance and inspection is vital to developing and maintaining critical infrastructure. Trimble's road infrastructure management workflow consists of the following six phases:

Phase 1, Field Data Collection. Trimble's MX60 and MX90 mobile mapping systems allow users to rapidly collect high-resolution, georeferenced imagery and light detection and ranging (LiDAR) data at highway speeds. These systems easily facilitate efficient surveying of extensive road networks without the need for road closures, thereby improving safety and reducing operational costs. The high-resolution LiDAR data supports AI-driven



workflows for road condition evaluation, facilitating more accurate and efficient assessments as well as enabling the detection and analysis of surface-level features such as potholes and wear patterns. A **pavement-facing camera** is a crucial component in Trimble's mobile mapping systems, particularly for applications involving crack detection.

Phase 2, Analysis/Feature Extraction. The collected data are processed using Trimble Business Center (TBC) office software, a comprehensive software package for civil engineers, surveyors and construction professionals that incorporates AI capabilities to automate tasks such as point cloud classification and feature extraction. This automation accelerates the generation of deliverables such as digital terrain models and road asset inventories, enhancing accuracy and efficiency.

Phase 3, Asset Lifecycle Management (ALM). Trimble's ALM approach, embodied in the **Trimble Unity** software suite, which leverages Esri's GIS innovations that span the entire lifecycle of infrastructure assets—from planning and design to construction, operation and maintenance. This comprehensive



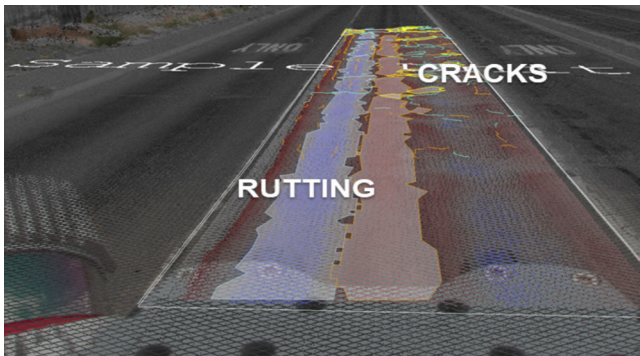
The Trimble mobile mapping systems have a combination of LiDAR and cameras, including a dedicated pavement camera, allowing detailed pavement analysis and mapping of the complete road corridor, capturing all assets.

framework enhances decision-making, optimizes resource allocation and reduces the total cost of ownership for transportation agencies.

Trimble Unity Maintain is the core enterprise asset management component within the Unity suite, providing transportation departments with tools to model, plan, schedule and execute maintenance tasks efficiently; use real-time data to allocate labor, equipment and materials effectively; and make informed decisions on asset rehabilitation and replacement. By integrating GIS data, Unity Maintain offers a spatially aware platform that improves visibility into asset conditions and maintenance needs. The suite also includes Pavement Analyst, which is designed for state and national transportation agencies managing extensive highway networks, and Pavement Express, which is tailored to local governments. Both Pavement Analyst and Pavement Express provide agencies the ability to take the current state of pavement assets (collected in phase 1) and model their deterioration through time as well as optimize treatments to ensure long-term performance. This provides agencies with the data needed to better maintain newly constructed assets and current assets.

Phase 4, Civil Construction and Design. This phase, primarily facilitated by TBC, involves converting raw survey and mapping data into detailed, constructible 3D models that are essential for guiding field operations, ensuring design accuracy and optimizing construction workflows. During design, contextual GIS information from the ALM solution can be brought into TBC using the same TBC GIS module.

Phase 5, Execution—Milling and Paving. Milling and paving is critical for translating digital designs into precise, high-quality road surfaces. This phase leverages Trimble's advanced machine



Trimble Business Center software offers AI-powered feature-extraction tools that can classify point clouds into various categories, including traffic signs, guard rails and road shoulders.

control systems: **Trimble Earthworks** and **Trimble Roadworks**.

Trimble Earthworks is an advanced grade control platform designed to enhance the accuracy and efficiency of earthmoving operations. It uses global navigation satellite system technology to guide heavy machinery according to 3D design models, ensuring precise cuts and fills. **Trimble Roadworks** is a 3D paving control platform that enhances the precision and efficiency of asphalt paving and milling operations.



Guided by the Trimble Roadworks Paving Control Platform for Asphalt Pavers, the paver precisely lays asphalt to grade and slope.

Phase 6, Maintenance. This phase is essential for evaluating the as-built conditions of roadways post-construction, ensuring that the constructed infrastructure aligns with design specifications and remains in optimal condition through time. The collected information also contributes to the “digital twin” demands of some owners. The metadata generated during construction and post-construction reality capture helps owners understand the true cost of ownership and shapes maintenance decisions in the future. This as-built survey is provided back to Trimble Unity (Phase 3), such that the maintenance team receives real-time data of the as-built conditions, ensuring a proper maintenance strategy is put in place for the pavement assets post-construction.

Return on Investment

Transportation departments around the world are using this workflow for both project delivery and asset management, increasing the return on their mobile mapping investments. For example, the California Department of Transportation (Caltrans) stated in its Caltrans Facts and SB1 Savings reports that the department saved nearly \$7 million in fiscal year 2021-2022 by using mobile mapping and automated machine guidance technology for several projects. Benefits resulted from changes in the way Caltrans provides survey data and how inspectors verify grades during rough grading as well as more efficient quantity estimation methods. During the full year, the agency saved nearly \$5 million in all earthworks projects and about \$1.9 million by reducing the number of construction working days. Such savings illustrate the tangible benefits of leveraging integrated digital solutions in road construction and maintenance.